Chapter 1 Master Study Guide — Introduction to the Science of Psychology

This is a complete walkthrough of Chapter 1 (textbook pages 1–39 plus the lecture slides), organized the way the chapter itself is organized, with every learning objective (LO 1–10), every key term, every named study, and the running twin story that ties the whole chapter together. It is written to be *explained*, not just skimmed — as if a student were talking through the entire chapter and assignment out loud.

The chapter has four big sections:

1. **What is psychology and how did it begin?** (definition, goals, history, perspectives)
2. **How do psychologists do research?** (the scientific method, research basics)
3. **Descriptive and correlational methods** (naturalistic observation, case studies, surveys, correlations)
4. **The experimental method** (cause and effect, variables, ethics, positive psychology)

THE STORY THAT RUNS THROUGH THE WHOLE CHAPTER: SAM AND ANAÏS

The textbook opens every concept with the true story of **Anaïs Bordier and Samantha Futerman**, and the professor's slides open with it too, so know the beats:

- **December 15, 2012:** Anaïs Bordier, a 25-year-old French fashion design student living in London (born in Busan, South Korea, adopted by Jacques and Patricia Bordier of Paris), sees a Facebook post of a YouTube video showing an American girl who looks EXACTLY like her — same face, same voice, even the same laugh.
- The look-alike turns out to be **Samantha Futerman**, an American actress in Los Angeles (also born in Busan, adopted by Judd and Jackie Futerman of New Jersey) — born **November 19, 1987, the exact same day as Anaïs**.
- According to both sets of adoption papers, each was a "single baby" — no twin mentioned. Anaïs sends Sam a Facebook message ("we looked really similar… like VERY REALLY SIMILAR"); Sam, stunned, accepts.
- They Skype for hours (their first call ran from about 12:30 A.M. to almost 4:00 A.M. Paris time), text daily on WhatsApp (inventing their own "pop" language), and discover a mountain of shared quirks: both

brush their teeth obsessively, both hate the feeling of shower curtains touching their skin, both “torpedo into a hole” when facing problems and feel better after a good nap.

- **May 14, 2013:** They meet in person in London — standing in a room “staring at each other, like two tiny dogs sniffing each other out,” then taking a nap together (“Not awkward at all. It took like two seconds.”).
- They crowdfund a documentary about their reunion called **Twinsters** on Kickstarter, which catches the attention of twin researcher **Dr. Nancy L. Segal** of California State University, Fullerton.
- Dr. Segal insists on a **DNA test** (“You need the biological proof”) — cheek swabs collected on Skype — which confirms they are **identical twins**.
- The twins participate in Segal’s research: tests of height, weight, hand preference, job satisfaction, self-esteem, personality traits, and cognitive abilities — all **variables**. The results show striking similarities and interesting differences: Anaïs scored higher on visual-spatial skills (fitting for a fashion designer), Sam scored better on some memory tests (fitting for an actress who memorizes lines and waited tables), and Sam is more extroverted (possibly because she grew up with two older brothers, while Anaïs was an only child).
- Their Korean birth mother, contacted through the adoption agency, denied ever having twins — the book explains the devastating stigma in South Korea around children born out of wedlock. The twins hold no resentment: “She gave them life, and she gave them each other.”
- The story ends in **positive psychology**: instead of mourning 25 lost years, they celebrate the reunion. Sam co-founded **Kindred**, a nonprofit foundation supporting adoptees and their families; Anaïs designs for the leather company Jean Rousseau and hopes to create an educational foundation for orphans and adopted children.

Why the story matters academically: identical twins raised apart are a natural experiment in **nature versus nurture** — same genes (nature), different environments (nurture). Their similarities point to nature; their differences point to nurture. Every research method in the chapter gets illustrated through twins.

SECTION 1: WHAT IS PSYCHOLOGY AND HOW DID IT BEGIN?

LO 1 — Describe the scope of psychology

Psychology is the scientific study of behavior and mental processes. Break the definition apart:

- **Behavior** = observable actions — gasping, smiling, laughing (what Anaïs did when she saw Sam's video).
- **Mental processes** = thoughts and emotions that are *not* directly observable — Anaïs wondering "Where was Samantha born? Was she adopted as well?" and feeling apprehension and excitement. Not observable, but still valid research topics.
- **Scientific** = psychologists are scientists; the claims come from systematic research, not intuition.

Psychologists are scientists who study behavior and mental processes. The field is broad: the **American Psychological Association (APA)** has over 50 divisions representing subdisciplines; the **Association for Psychological Science (APS)** lists over 100 affiliated societies and organizations. Each chapter of the textbook is a subfield.

Two major types of research:

- **Basic research** — collecting data to support or refute theories; gathering knowledge **for the sake of knowledge**; often occurs in university laboratories. Examples: general explorations of human memory, sensory abilities, responses to trauma.
- **Applied research** — focused on **changing behaviors and outcomes**; often leads to real-world applications, and is often conducted in natural settings outside the laboratory. Examples: behavioral interventions for children with autism, keyboard layouts that improve typing performance. (Applied research may build on findings from basic research.)

Test-prep example from the book: experts using prior findings to design a campaign to reduce student binge drinking = **applied research**.

LO 2 — The four goals of psychology (Describe, Explain, Predict, Change)

The chapter's example: a psychologist studying **mindfulness meditation in elementary schools**.

1. **DESCRIBE** — observe and report what is observed. She assesses students' academic performance, social adjustment, and physical health before and during a meditation program, then publishes her observations.
2. **EXPLAIN** — organize and make sense of observations. Students get fewer detentions and suspensions after the program starts; she searches the literature (e.g., behavior changes in meditating assisted-living residents) to explain why.
3. **PREDICT** — predict behaviors or outcomes on the basis of observed patterns. If meditation reduced detentions here, she predicts

meditation in another setting (like a correctional institution) might lead to the same outcome.
4. **CHANGE** — use research findings to modify behavior; *apply* findings in a beneficial way — helping schools implement meditation in their curricula.

Real-world sidebar from the book: a Baltimore elementary school replaced detentions with a “Mindful Moment Room” — in the three years since, not a single student was suspended (versus four suspensions the prior year); a small study of sixth graders found meditation practice was associated with reduced risk of suicidal ideation.

Mnemonic: **D-E-P-C** — “**D**oes **E**veryone **P**ractice **C**almly?” Describe → Explain → Predict → Change.

LO 3 — Psychology’s roots (covered in depth in the People to Know document)

The compressed timeline:

- **Philosophy era:** Plato (innate knowledge — nature) → Aristotle (empiricism, learning through senses — nurture) → Descartes (dualism: body is a machine, mind is non-physical; “I think, therefore I am”) → [Locke: blank slate — from standard history].
- **Nature** = inherited biological factors shaping behavior, personality, characteristics. **Nurture** = environmental factors doing the same. Modern psychology’s answer: **both**, studied through twin research. Identical twins share 100% of their genes; fraternal twins (like non-twin siblings) share about 50%. Identical twins raised apart are “equal in nature but not in nurture” — so their similarities implicate nature, their differences implicate nurture. Identical twins even raised apart tend to be very close in measured intelligence (Bouchard et al., 1990), suggesting genes play a major role in intellectual ability. The butterfly example: two butterflies of the same species look wildly different because they were born in different seasons — pure nurture (environment) at work.
- **1879 — psychology is born:** Wilhelm Wundt founds the first psychology laboratory at the University of Leipzig, Germany. “There were no psychologists until 1879.” Wundt = “father of psychology,” method = **introspection** (examination of one’s own conscious activities), demanding objectivity and 10,000 practice observations; his 1861 pendulum-and-bell experiment showed mental processes take measurable time (~1/10 second).
- **Early schools:** Titchener’s **structuralism** (introspection to find the structure and most basic elements — “atoms” — of the mind; Cornell 1893; subjective reports; did not outlast his lifetime) versus James’s **functionalism** (the *function* of thoughts, feelings, behaviors — how

they help us adapt; inspired by Darwin; consciousness as an ever-changing “stream”; influenced educational psychology, emotion research, comparative animal studies).

- **Women who broke through the ceiling:** Calkins (denied Harvard PhD, first female APA president 1905), Washburn (first woman psychology PhD, Cornell 1894, *The Animal Mind*), Mamie Phipps Clark (first Black woman psychology PhD from Columbia, prejudice/discrimination research, Brown v. Board 1954, Northside Center). Today women earn ~71% of advanced psychology degrees — the slides ask: might *this* imbalance be problematic too?

LO 4 — The major perspectives in psychology (Table 1.1 — memorize this)

Early schools (structuralism, functionalism) faded, but they seeded today’s **eight perspectives**. For each: main idea + the question it asks + names attached.

1. **Psychoanalytic** (Freud): Underlying **unconscious conflicts** influence behavior. *How do unconscious conflicts affect decisions and behavior?* Early childhood matters; talk therapy; the “Freud Problem” (famous but lacking scientific support).
2. **Behavioral** (Pavlov → Watson → Skinner): Behavior is **learned** through associations, reinforcers, and observation. *How does learning shape behavior?* Strong nurture position. Classical conditioning (Pavlov), behaviorism as a school (Watson), operant conditioning (Skinner).
3. **Humanistic** (Rogers & Maslow): Humans are naturally inclined to **grow in a positive direction**. *How do choice and self-determination influence behavior?* A rebellion against the determinism of Freud and the behaviorists.
4. **Cognitive** (George Miller and the 1950s “cognitive revolution”): Behavior is driven by **cognitive/mental processes** — thinking, memory, language. *How do thinking, memory, and language direct behavior?* During behaviorism’s prime (1930–1950), mental processes were off-limits; the 1950s brought them back. Miller’s memory research (his famous “7 plus or minus 2” capacity finding, per the slides) was a catalyst. New branch: **cognitive neuroscience** — physiological explanations for mental processes, connecting behavior to the nervous system, especially the brain (flourishing thanks to brain-scanning technology).
5. **Evolutionary** (Darwin’s theory; David Buss as modern founder): Behaviors and mental processes are shaped by **natural selection** — inherited traits increase in frequency when adaptive, decrease when maladaptive. *How does natural selection influence thoughts,*

emotions, and behaviors? Finch-beak example; explains personality traits, intelligence, risk-taking.

6. **Biological**: Behavior and mental processes arise from **physiological activity** — hormones, genes, brain activity. *How do biological factors influence behavior and mental processes?* Overlaps with **neuroscience** (study of the brain and nervous system).
7. **Sociocultural** (Lev Vygotsky, 1896–1934, Russian psychologist): **Other people and the broader cultural context** influence behavior and mental processes. *How do interactions with other people and cultural factors shape behaviors and attitudes?* Vygotsky proposed that parents, teachers, and peers play a critical role in a child's cognitive development. In the 1980s, cross-cultural research revealed that Western research participants are not always representative of people from other cultures — and even groups *within* a culture differ. (Book example: bargaining is expected in many Asian markets but not in fixed-price US stores — cultural context matters.)
8. **Biopsychosocial**: Behavior is explained through the **interaction of biological, psychological, and sociocultural factors** — a combined lens used by scientists in many fields (from mental-health researchers to physicians treating sickle-cell disease). *How do the interactions of biology, psychology, and culture influence thoughts, emotions, and behaviors?*

How similar/different (a "Show What You Know" question): sociocultural and biopsychosocial are similar in examining how interactions with others influence us; cognitive versus biological differ in that cognitive focuses on thought processes while biological emphasizes physiological processes.

Human behavior is complex — it often "requires an integrated approach, using the findings of multiple perspectives." Psychologists pick and choose among approaches; theoretical **models** help clarify complex observations.

Practice like the slides ask: explain *drug use* through four perspectives — e.g., psychoanalytic (unconscious conflict soothed by substances), behavioral (drug use reinforced by pleasurable consequences), biological (genetic predispositions, brain chemistry), sociocultural (peer and cultural norms around use).

SECTION 2: HOW DO PSYCHOLOGISTS DO RESEARCH?

LO 5 — The scientific method

Scientific method = the process scientists use to conduct research, which includes a continuing cycle of exploration, critical thinking, and systematic observation. Goal: **empirical evidence** — data from systematic observations or experiments, used to support or refute a hypothesis.

Observations must be **objective** (outside the influence of personal opinion and expectation) — humans are prone to errors in thinking, and the method minimizes their impact.

Key terms first:

- **Hypothesis** = a statement that can be used to test a prediction.
- **Theory** = synthesizes observations in order to explain phenomena and guide predictions to be tested through research. A theory is NOT a guess or hunch — it is “a well-established body of principles that rest on a sturdy foundation of scientific evidence.” (Evolution is the book’s example of a theory mistaken by the public for an “ongoing controversy” — in reality it’s supported by the overwhelming majority of scientists.)
- **Experiment** = a controlled procedure involving careful examination through scientific observation and/or manipulation of **variables** (measurable characteristics).

The five steps, illustrated twice — by Dr. Nancy Segal’s twin study and by the Infographic 1.1 reading-comprehension study:

1. **Develop a question.** Comes from observing the world, personal interest, and reviewing the scientific literature. Segal watched fraternal twins fight over a puzzle at a birthday party and wondered: would identical twins cooperate better than fraternal twins? Infographic example: “We’re constantly interrupted by calls and people at the door — what does that do to reading comprehension?”
2. **Develop a hypothesis.** Segal’s: *When given a joint task, identical twins will cooperate more and compete less than fraternal twins* (rooted in behavioral genetics and evolutionary theory). Infographic’s: *Interruptions while reading will lead to poorer comprehension.* Researchers look for existing theories and establish operational definitions.
3. **Design study and collect data.** Segal videotaped identical and fraternal twin children solving a puzzle together, then rated “indices of cooperative behavior” — how often they handed each other pieces, leaned on/pushed/hit one another, even counted facial expressions (sadness, surprise, pride). **Operational definition** = the precise manner in which a variable of interest is defined and measured (Segal operationally defined cooperation via how often twins worked together, accepted help, smiled at each other). Data collection must be controlled — e.g., a standardized, automated personality test is more objective than casual impressions. Infographic example: participants read SAT passages in an interruption condition (math problems between paragraphs) vs. no-interruption condition, then took comprehension tests.

4. **Analyze the data.** Raw data are meaningless until analyzed. **Descriptive statistics** organize and present data (tables, graphs, charts); **inferential statistics** go beyond describing — they let researchers make inferences and determine the probability of events. Did results support the hypothesis? Segal's did: "The identical twins were more cooperative on almost every index that I used… identical genes do contribute to the greater cooperation observed between partners." Infographic finding: interrupted readers comprehended worse — *unless* the interruption was preceded by a 15-second break.
5. **Publish the findings.** Submit to a scholarly, **peer-reviewed** journal (subject-matter experts scrutinize, recommend publish/revise/reject) or present at conferences. Publication enables **replication** = repeating an experiment, generally with a new sample and/or changes to procedures, to provide further support for the findings. The more a study is replicated with similar findings, the more confidence we can have.

Why replication and peer review matter — the vaccine-autism fraud: In the late 1990s, Wakefield et al. (1998) published a study claiming vaccines caused autism. The study was **fraudulent** (fabricated/deceptive findings), but it took *12 years* for The Lancet to retract it. Researchers tried for over 10 years to replicate the result and never could; multiple high-quality studies found no credible support for a vaccine–autism link (CDC agrees: no link). But the damage persists — some parents still refuse vaccines, and measles outbreaks among unvaccinated children have occurred in the US. Lessons: peer review isn't foolproof (about one article a day is retracted — plagiarism, data meddling; retractions happen more in top journals), replication is the safety net, and bad science can have real public-health consequences.

Ask new questions: most studies generate more questions than answers — the scientific method is **cyclical** (the infographic's researchers ended by wondering whether *experts* get interrupted with less damage — a new study is born).

Critical thinking = the process of weighing various pieces of evidence, synthesizing them, and evaluating and determining the contributions of each; disciplined thinking that is clear, rational, open-minded, and informed by evidence. A critical thinker is skeptical, thinks deeply, evaluates claims using existing knowledge, asks questions, considers alternative explanations, reflects on their own emotional reactions, tolerates uncertainty, stays open-minded. Infographic 1.2's critical questions when reading about research: Who wrote it (professional background)? Where was it published (peer-reviewed vs. popular press)? What are the findings (limitations cited? other variables?)? What methods were used (sample size? data collection)? Has it been replicated? (Framing example: 97% of leading climate scientists agree human activity drives warming — yet many people underestimate the problem; critical thinking applies to everyday dilemmas and global crises alike.)

LO 6 — Research basics: variables, population, and sample

- **Variables** = measurable characteristics that can vary over time or across people (in psychology: reaction times, problem-solving approaches, aggressive behaviors, job satisfaction, self-esteem, personality traits, cognitive abilities — the exact battery Segal ran on Sam and Anaïs).
- **Population** = ALL members of an identified group about which a researcher is interested (e.g., all US college students).
- **Sample** = the subset of the population actually included in the study (like a blood test — the lab doesn't take all your blood).
- **Random sample** = a subset chosen through a procedure that ensures **every member of the population has an equal chance of being selected** (e.g., randomly selecting high-school seniors from SAT/ACT databases). Purpose: avoid selection bias at the *recruiting* stage.
- **Representative sample** = a subgroup whose **characteristics are similar to those of the population of interest**. Random sampling increases the likelihood of getting a representative sample. Why it matters: only a representative sample lets you **generalize** findings to the larger population. Book example: studying US attitudes toward undocumented workers but recruiting only in Los Angeles (one of the three biggest undocumented-immigrant populations, with New York and Houston) biases the result — LA residents don't represent the whole country. If 44% of a truly representative sample holds a view, you can infer "approximately 44% of people in the United States" do.
- Be careful making inferences from **small samples** to large populations.

SECTION 3: DESCRIPTIVE AND CORRELATIONAL METHODS

LO 7 — Descriptive research

Descriptive research = research methods that describe and explore behaviors, but with findings that **cannot definitively state cause-and-effect relationships**. Best for **new or unexplored topics** where researchers might not have specific expectations about outcomes. Per the slides, four major types: naturalistic observation, case study, survey method, and correlations.

1. Naturalistic observation = studying participants in their **natural environments** through systematic observation. "Natural" doesn't mean wilderness — an office, a preschool, a dorm, a family home.

- Book example: researchers videotaped 30 couples at home; husbands were more likely to *give* support (usually practical, like getting kids ready for school, rather than emotional), wives more inclined to *ask* for help — likely because they were "taxed with an objectively greater share of the domestic workload."
- Hand-washing study: observers sat quietly in campus bathroom stalls with stopwatches; most men and women washed after using the toilet, but almost **half of men** failed to wash after using urinals. (Would they have washed if they knew they were watched? That's the point of hiding the observers.)
- Golden rule: **do not disturb the participants or their environment** — otherwise they change their normal behavior.
- Challenges: variables must still be operationally defined (defining "aggression" in toddlers requires a detailed checklist of screeching, pushing, etc., plus a coding system); natural environments are cluttered with unwanted variables you can't remove; replication is hard; you can't control who shows up (whoever joins the Lego table becomes a participant).
- **Observer bias** = errors in the recording of observations, the result of a researcher's value system, expectations, or attitudes. Fix: use **multiple observers** and compare how similarly they record the same behaviors.

2. Case study = a detailed examination of an individual or small group, using multiple avenues to gather information (in-depth interviews with the person plus friends, family, coworkers; questionnaires on medical history, career, mental health).

- Invaluable for **rare events** — like identical twins born in South Korea and reared on different continents, or an astronaut's health compared to his identical twin on Earth.
- The **"Jim Twins"**: identical twins Jim Springer and Jim Lewis, separated shortly after birth, reunited at age 39. Both named "James" by their adoptive parents; both gravitated to math and carpentry as kids; each had a dog named "Toy," a first wife named "Linda," and a second wife named "Betty"; both smoked Salems, drove the same type of blue Chevy, and vacationed at the same Florida spot.
- Limits: a case study is **a sample of one** — it cannot be used to support or refute a hypothesis (hypothesis testing requires comparisons between conditions), and you must not make sweeping generalizations from a single person or family. It's useful for guiding the design of studies on underexplored topics and furthering theory development. Unlike the detached naturalistic observer, the case-study researcher may be completely immersed in the participant's environment (which can affect objectivity).

3. Survey method = descriptive research using **questionnaires or interviews** to gather data (paper, face-to-face, smartphone/tablet). Benefit: data from numerous people in a short time. Four cautions:

- **Wording effects / response bias:** "global warming" versus "climate change" — half a representative sample was asked whether *global warming* will harm their family, half got the identical question with *climate change*; "global warming" set off more alarm bells across every group surveyed (Figure 1.2). Positive/negative spin sways answers (half-full vs. half-empty).
- **Honesty:** people lie, especially face-to-face on sensitive topics. In one study, men and women reported relationship cheating equally when hooked to a (fake) lie detector, but men were more honest than women when reports were anonymous. The bias toward **social desirability** (answering to look good) means researchers recommend collecting sensitive information anonymously.
- **Skimming the surface:** surveys capture *what* people say, not *why*; a "yes, I intend to exercise" can mean anything from vague wish to concrete plan. More precise: scaled responses (5-point strongly agree → strongly disagree; never → almost always).
- **Sampling problem / response rates:** send 100 surveys, get 20 back — are those 20 representative of the group? Low response rates threaten representativeness.
- Survey in action — "Digital Hypocrisy" (Scientific American): national survey of ~1,800 parents of kids 8–18. Parents average **9 hours 22 minutes** of daily screen time, nearly 8 hours of it personal (not work) — yet **78%** believe they're good technology role models. Kids feel ignored and mimic parents; tween/teen technology use correlates with shorter attention spans and worse school performance; infants struggle with emotional/nonverbal cues when parents are buried in phones (the "still face phenomenon"). Good news: 94% of parents recognize technology can support education; shared screen activities enhance learning; set limits for kids *and* yourself (device-free dinner table).

LO 8 — The correlational method

Correlational method = a type of descriptive research examining the **relationships among variables**, helping researchers make predictions. A **correlation** = an association or relationship between two (or more) variables.

- **Positive correlation:** variables move in the same direction — more parental reading time, bigger child vocabulary; more study hours, higher grades.

- **Negative correlation:** as one goes up the other goes down (inverse) — more smartphone time, lower academic performance; more shopping, less money in the bank.
- **Correlation coefficient (r)** = the statistical measure that indicates the **strength and direction** of the relationship. Range: **–1.00 to +1.00**. Sign = direction (positive/negative). Distance from zero = strength: close to ±1.00 is strong; close to 0.00 is weak/none. So r = +.73 is a strong positive correlation (e.g., study hours and grades); r = .15 (salary and job satisfaction, across 92 studies) is weak — "salary isn't everything"; shoe size and intelligence in adults ≈ 0. Perfect +1.00 and –1.00 lines, strong scatters, and no-relationship clouds are shown on **scatterplots** (each dot = one participant's scores on the two variables).
- **CORRELATION DOES NOT PROVE CAUSATION.** Two reasons:
  - **Third variable** = some unaccounted-for characteristic of the participants or their environment that explains changes in both variables. Kids' shoe size correlates with reading scores — the third variable is age/development. Studying time correlates with grades — but class attendance drives both. Violent-media exposure correlates with aggression — but *parenting style* (limiting/monitoring/discussing media) influences both.
  - **Directionality:** even if there's a causal link, which way does it run? Does violent media make children aggressive, or are aggressive children drawn to violent media? "The causal direction can go both ways."
- Why bother with correlational research at all? It provides clues about what underlies behaviors and enables prediction, and it's essential **when experiments would be unethical or impossible** (you can't ethically manipulate children's exposure to violence).
- **Across the World — the happiest places on the planet:** the World Happiness Report (UN Sustainable Development Solutions Network) finds Earth's happiest people in **Finland**; runners-up Norway, Denmark, Iceland; the **US ranks 18th of 156**; unhappiest spots include several African nations, Syria, Yemen, Haiti. Key correlates of happiness: **social support** — deep, meaningful relationships (Latin American countries score high on happiness despite poverty because of "close, warm, and genuine interpersonal relations" — "there is more to life than income"), and **sustainable development** (meeting present needs without compromising future generations — economic progress balanced with environmental and social protection). Governments now use happiness data in policy decisions. This is a great example of *descriptive* research guiding real decisions.

Media literacy application ("Don't believe everything you read"): 2017 headlines screamed "Tablets and smartphones damage toddlers' speech development.” The actual research: parents of ~1,000 children (6–24 months) *reported* screen time and filled out language questionnaires — researchers found **a link (correlation)** between screen time and speech problems. The headlines implied causation. Alternative explanations: a third variable (parents who interact/talk less may both allow more screens and provide less speech stimulation) or reverse directionality (speech-delayed children may be more drawn to digital entertainment). Take-home: **if a news report claims X causes Y, check whether the research was correlational.** Related media trap: **“false balance”** — presenting a 97%-versus-3% scientific consensus (like human-caused climate change) as a 50-50 “debate” between two talking heads.

SECTION 4: THE EXPERIMENTAL METHOD

LO 9 — How experiments establish cause and effect

Experimental method = a type of research that **manipulates a variable of interest (the independent variable) to uncover cause-and-effect relationships**. The logic: make two or more groups **equivalent in every way except one** — the treatment. If the groups then differ on the outcome, the manipulation caused the difference.

Running example: does moderate caffeine improve attention? Hypothesis: *participants given a moderate dose of caffeine will perform better on a task requiring increased attention compared to participants given a sugar pill.*

The machinery, term by term:

- **Random assignment** = the process of appointing study participants to the experimental or control groups, ensuring every person has an **equal chance of being assigned to either** (coin flip, dice, computer). Purpose: prevent lopsided groups (all teenagers in one group, all middle-aged in the other would wreck an attention study). **Do not confuse with random sampling:** random *sampling* happens at the start, choosing participants from the population; random *assignment* happens later, dividing participants into groups.
- **Experimental group** = the members exposed to the treatment/manipulation (real caffeine) — the treatment group.
- **Control group** = participants NOT exposed to the treatment variable (identical-looking sugar pill) — the comparison group. (Cartoon on the slides: lab rats watching agitated neighbors — “Well, I guess we’re the control group.”)
- **Independent variable (IV)** = the variable the researcher **deliberately manipulates** (caffeine vs. sugar pill). There can be more than one IV. [Synonym: explanatory variable.]

- **Dependent variable (DV)** = the characteristic or response that is **measured** to determine the effect of the manipulation (performance on attention tasks). The DV “depends” on the IV. [Synonym: response variable.]
- **Extraneous variable** = a characteristic of participants or the environment that could **unexpectedly influence** the outcome (three participants are so caffeine-sensitive they can’t sleep — sleep deprivation now affects attention scores).
- **Confounding variable** = a type of extraneous variable that **changes in sync with the independent variable**, making it impossible to tell which variable caused the change in the DV (testing the experimental group in the morning and the control group in the afternoon confounds caffeine with time of day — and research suggests caffeine’s impact depends on time of day).
- **Controlling variables:** random assignment helps balance groups; treat experimental and control groups *exactly the same* (same time, same location); or eliminate the problem via participant selection (screen out the caffeine-sensitive).
- **Placebo** = an inert substance given to members of the control group; the fake treatment has no benefit but is administered as if it did. The **placebo effect**: “thinking is believing” — people given fake treatments often improve because they *expect* to; placebos can ease pain, anxiety, depression, even Parkinson’s symptoms, and the effect strengthens when the person delivering the placebo shows warmth and competence. (Placebos can’t shrink tumors, but they can help treatment side effects like pain, fatigue, nausea.)
- **Single-blind study:** participants don’t know which treatment they’re receiving.
- **Double-blind study = neither the researchers administering the IV nor the participants** know who is getting the treatment and who is getting the placebo. Manages participant expectations AND experimenter influence. Deception here is a legitimate scientific tool (revealed later in debriefing).
- **Experimenter bias** = researchers’ expectations that influence the outcome of a study [synonyms: experimenter effect, researcher expectancy effect] — subtle verbal/nonverbal cues (“I really have high hopes for this medicine”), or the way a researcher frames questions and interprets findings. The classic example: **Clever Hans**, the horse who “counted” and solved math problems by tapping his hoof. Hans was actually a brilliant reader of *human* behavior — he could tell from the questioner’s involuntary cues when to stop tapping. If the questioner didn’t know the answer, or Hans wore blinders, the ability vanished. Moral: expectations leak; blind the humans.

Strengths and weaknesses of the experimental method: it stands alone in establishing cause and effect, with control over participants, variables, and implementation — but laboratory settings are inherently unnatural (behavior changes when people know they're observed), labs are expensive, and data collection can be slow (Table 1.2 summary below).

Experiment in action — "SpongeBob on the Brain": sixty 4-year-olds (mostly White, upper-middle-class — note the representativeness limitation) were **randomly assigned** to one of three conditions: watch 9 minutes of *SpongeBob SquarePants* (extremely fast-paced), watch 9 minutes of a slow educational program, or draw with crayons and markers for 9 minutes. Then all took four tests of **executive function** (self-control, decision making, problem solving, and other higher-level functions). Result: the SpongeBob group performed considerably worse — just 9 minutes produced a temporary lapse in cognitive function. IV = the 9-minute activity; DV = performance on the cognitive tests. Preexisting attentional issues and home TV habits were accounted for. Follow-up research suggests the culprit is "fantastical" content (characters magically changing shape, popping in and out of existence) — even slow-paced fantastical shows (*Little Einsteins*) compromise short-term executive function. Critical-thinking caveats: results were temporary, and the sample limits generalization.

Table 1.2 — Research methods: advantages and drawbacks

Method	Advantages	Disadvantages
Descriptive	Good for tackling new research questions and studying phenomena in natural environments	Very little control; increased experimenter/participant bias; cannot determine cause and effect
Correlational	Shows whether two variables are related; useful when an experiment isn't possible	Directionality and third-variable problems; cannot determine cause and effect
Experimental	Can determine cause and effect; increased control over variables	Results may not generalize beyond the lab; potential for extraneous variables

How to choose? **It depends on the research question.**

LO 10 — Research ethics

Psychologists study humans and other living creatures "who experience pain, fear, and other complex feelings" — treating them with dignity and respect is a professional duty. Professional organizations — **APA, APS, and the British Psychological Society (BPS)** — publish written guidelines members agree to follow. The encouraged principles:

- **Do no harm.**
- **Safeguard the welfare of humans and animals** in research.
- **Know responsibilities** to society and community.
- **Maintain accuracy** in research, teaching, and practice.
- **Respect human dignity.**

(Historical studies you'll meet in Chapter 11 would never be approved today.) Notions of "ethical treatment" vary — not everyone agrees with these codes.

The working parts:

- **Institutional Review Board (IRB)** = a committee that reviews research proposals to protect the rights and welfare of all participants. **All experiments on humans and animals must be approved by an IRB.** [Synonyms: reviewing committee, Animal Welfare Committee, Independent Ethics Committee, Ethical Review Board.]
- **Informed consent** = acknowledgment from study participants that they understand what their participation will entail — including any possible harm. Dr. Segal describes it as "kind of like your contract with them": a letter laying out what participants will do, benefits, risks, confidentiality guarantees — without giving away hypotheses (to avoid biasing findings). Participation is **completely voluntary**; participants may refuse any task or **drop out at any time**. Research on minors requires consent from parents/guardians.
- **Debriefing** = sharing information with participants after their involvement has ended, including the purpose of the research and **any deception used** (e.g., revealing who actually got the placebo in a double-blind study). Deception is allowed when necessary but must be disclosed afterward.
- **Confidentiality:** researchers must protect data from misuse or theft; therapists must keep client and session information confidential so clients can speak freely. Legal exceptions where confidentiality **must** be broken: the client is a danger to self or others; cases of abuse, neglect, or domestic violence; court orders. Psychologists may also share limited information for payment purposes ("minimum that is necessary") or at the client's request.

POSITIVE PSYCHOLOGY (the chapter's closing idea)

Positive psychology = an approach that focuses on the positive aspects of human beings, seeking to understand their strengths and uncover the roots of happiness, creativity, humor, love — "all that is best about people" (Seligman & Csikszentmihalyi, 2000). Historically psychology focused on the abnormal and maladaptive; positive psychology doesn't deny the dark side — "it just directs the spotlight elsewhere." Positive psychologists believe humans "strive to lead meaningful, happy, and good lives." It is similar to the **humanistic perspective**, whose early work set the stage for it. Evidence: people with a positive outlook tend to have better mental and physical health than less optimistic peers. Chapter example: Sam and Anaïs — instead of lamenting 25 lost years, they rejoice in reunion, anticipate good things, and help other twins and adoptees (Kindred foundation; planned educational foundation for orphans and adoptees).

SUMMARY OF LEARNING OBJECTIVES (LO 1–10, one breath each)

1. **Scope of psychology:** the scientific study of behavior and mental processes; basic research gathers knowledge for its own sake, applied research changes behaviors and outcomes.
2. **Goals:** describe, explain, predict, change.
3. **Founders:** philosophy + physiology roots; nature–nurture debate; 1879 Wundt's Leipzig lab; Titchener's structuralism; James's functionalism (first US classes); Calkins, Washburn, Clark broke barriers.
4. **Perspectives:** psychoanalytic, behavioral, humanistic, cognitive, evolutionary, biological, sociocultural, biopsychosocial — different vantage points on complex behavior.
5. **Scientific method:** develop a question → hypothesis → design study & collect data → analyze → publish; cyclical, with critical thinking at every step.
6. **Random vs. representative sample:** random = everyone has an equal chance of selection; representative = sample characteristics mirror the population; random sampling promotes representativeness, which permits generalization.
7. **Descriptive research:** naturalistic observation, case studies, surveys — describe and explore, especially new topics; cannot establish cause and effect.
8. **Correlational method:** measures relationships (r from –1.00 to +1.00; sign = direction, magnitude = strength); enables prediction but NEVER proves causation (third variables, directionality).

9. **Experimental method:** manipulates the IV, measures the DV, holds everything else constant via random assignment and controls; the one method that can determine cause and effect.
10. **Ethics:** do no harm; IRB approval; informed consent; debriefing; confidentiality; welfare of humans and animals; accuracy and dignity.

COMPLETE KEY TERMS CHECKLIST (from the chapter's own list)

behavioral perspective · behaviorism · biological perspective · biopsychosocial perspective · case study · cognitive perspective · confounding variable · control group · correlation · correlation coefficient · correlational method · critical thinking · debriefing · dependent variable (DV) · descriptive research · double-blind study · evolutionary perspective · experiment · experimental group · experimental method · experimenter bias · extraneous variable · functionalism · humanistic psychology · hypothesis · independent variable (IV) · informed consent · Institutional Review Board (IRB) · introspection · natural selection · naturalistic observation · nature · nurture · observer bias · operational definition · placebo · population · positive psychology · psychoanalytic perspective · psychologists · psychology · random assignment · random sample · replicate · representative sample · sample · scientific method · sociocultural perspective · structuralism · survey method · theory · third variable · variables

(All of these are defined in this guide and/or the glossary document.)